

Mathematics 629
Spring 2026
Homework Assignment 6

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Due: 3/25/2026

1. Write out the complete details of the proof of the “differentiation under the integral” Theorem 1.129 in the notes.

Proof.

□

2. Prove that $\int_{(0,\infty)} x^n e^{-x} dx = n!$ by differentiating $\int_0^\infty e^{-tx} dt = 1/x$.

Proof.

□

3. Prove that if f_n is a sequence of decreasing nonnegative measurable functions with $\int f_1 d\mu < \infty$, then

$$\int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Proof.

□

4. Let $y_1, \dots, y_M$ be distinct points in $\mathbb{R}$, and $\beta_1, \dots, \beta_M \in \mathbb{R}$. Let

$$f(x) = \prod_{j=1}^M |x - y_j|^{-\beta_j}.$$

Find a necessary and sufficient condition on $(\beta_1, \dots, \beta_M)$ for f to belong to $L^1(\mathbb{R}, m)$.

Proof.

□

5. Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} |x|^{-1/3}, & |x| < 1, \\ 0, & |x| > 1. \end{cases}$$

Let $\{r_k\}_{k=0}^\infty$ be an enumeration of all points in $[0, 1] \cap \mathbb{Q}$. Let

$$f(x) = \mathbf{1}_{[0,1]} \sum_{k \in \mathbb{N}} 2^{-k} g(x - r_k).$$

(i) Show that $f \in L^1([0, 1])$.

Proof.

□

(ii) Show that f is unbounded on every open set (and remains so after modification on a set of measure zero). Show that f is discontinuous at every point in $[0, 1]$ and remains so after modification on a set of measure zero.

Proof.

□

(iii) Let $0 < \varepsilon < 1$. Construct a compact subset K_ε of $[0, 1]$ such that $m(K_\varepsilon) \geq 1 - \varepsilon$ and $f\mathbf{1}_{K_\varepsilon}$ is continuous on K_ε .

Proof.

□

6. Let $f \in L^1([0, 2\pi])$.

(i) Prove that $\lim_{n \rightarrow \infty} \int_0^{2\pi} f(x) e^{-inx} dx = 0$.

Proof.

□

(ii) Prove that for a set $E \subset [0, 2\pi]$ of positive Lebesgue measure $m(E)$,

$$\lim_{n \rightarrow \infty} \int_E \cos^2(nx + \gamma_n) dx = \frac{m(E)}{2},$$

for any sequence γ_n .

Proof.

□

(iii) Let $F_n(x) = a_n \cos nx + b_n \sin nx$. Assume that the series $\sum_{n=0}^{\infty} F_n(x)$ converges on a set E of positive measure. Show that $\lim_{n \rightarrow \infty} a_n = 0$ and $\lim_{n \rightarrow \infty} b_n = 0$.

Hint: It might help to express $F_n(x)^2$ in the form $(r_n \cos(nx + \gamma_n))^2$, with $r_n > 0$.

Proof.

□